

Network IDS

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Objective

To develop a **basic prototype of a Network Intrusion Detection System (NIDS)** that monitors live network traffic and detects suspicious activities (e.g., port scans, unauthorized access attempts) using signature-based detection techniques.

This PoC aims to:

- Demonstrate packet sniffing capabilities
- Detect basic network threats in real time
- Lay the foundation for an extensible and scalable IDS

Use Cases

- Real-time network monitoring in small environments (e.g., labs, startups)
- Security education and training
- Early-stage prototype for more advanced IDS/IPS systems
- Detection of:

Port scans (SYN scan detection)

Unauthorized access to sensitive services (e.g., SSH, RDP)

Suspicious traffic patterns

Tools Used

- **Python** – Programming language for implementation
- **Scapy** – Powerful Python library used for packet sniffing, decoding, and manipulation

- **Terminal / CLI** – To run and test the PoC
- **Wireshark (optional)** – For validating packet capture behavior
- **Logging module (optional)** – For saving alerts to a log file

What does it use?

- **Python** – Simple and powerful programming language
- **Scapy** – A Python library that lets you read and analyze network packets
- **Command Line (Terminal)** – To run the script and view alerts

Detection Logic

Detection is signature-based. Each suspicious activity is defined by a simple rule, such as a packet with the SYN flag only, or one that targets specific TCP ports. While these rules are hardcoded in the PoC, they could easily be externalized into a JSON or YAML configuration for easier management in a production version.

How does it work?

1. You run the script.
It starts “sniffing” your network traffic (like listening to everything going in and out).
2. It looks at TCP packets.
These are the basic building blocks of internet communication — used for websites, apps, remote access, etc.
3. It checks for danger signs.
For example:
 - A packet going to port 23 (Telnet — often targeted by hackers)
 - A packet with only the SYN flag (which might be a scan, like someone checking if a port is open)
 - Access to dangerous ports like 3389 (RDP), 22 (SSH), or 4444 (backdoors)

4. It prints an alert if it finds something weird.
It shows you the IP address, port, and time — like this:

What you need to run it

1. Install scapy:
`pip install scapy`
2. Run the script with admin access:

```
sudo python3 ids_poc.py
```

Sample Output

- [2025-08-18 10:22:10] ALERT: SYN scan detected:
192.168.1.10 -> 192.168.1.1:80
- [2025-08-18 10:22:12] ALERT: Connection to
suspicious port 23: 192.168.1.12 -> 192.168.1.1